

Exp no:10 Implement Boundary condition between two different dielectric medium in Magnetic Field and find the relative permeability of the different medium.

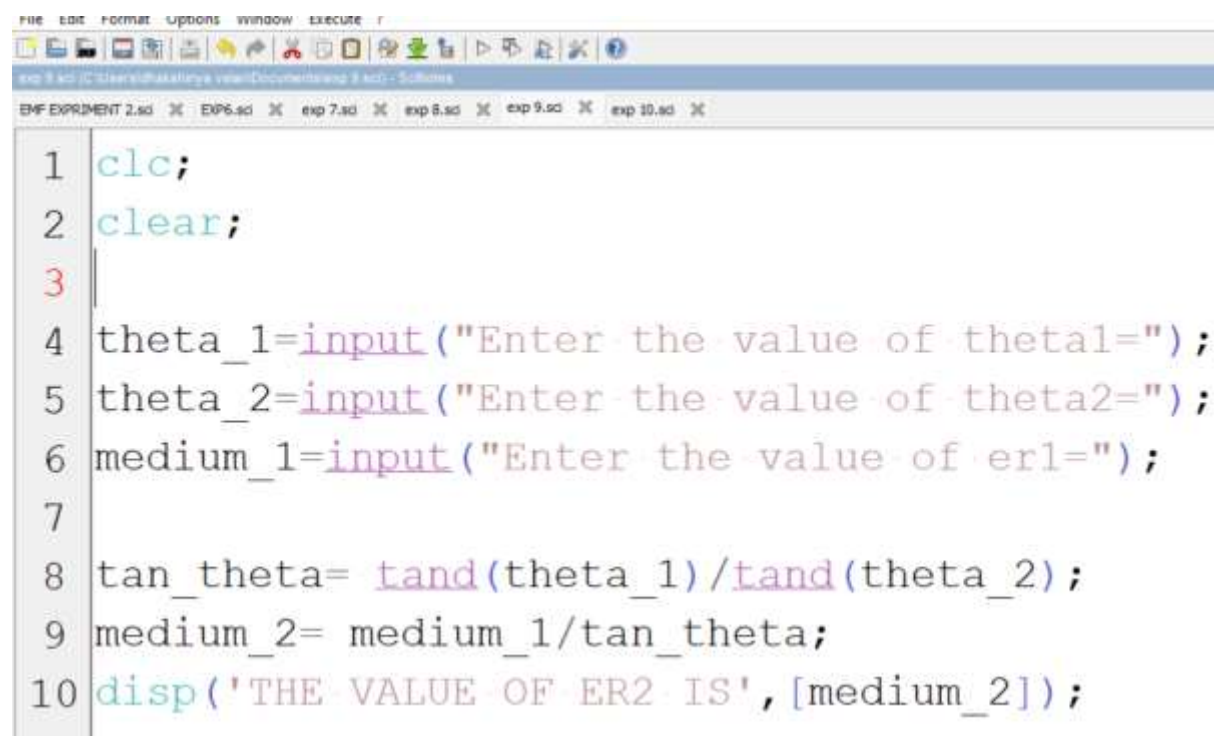
Aim:

To write program for Boundary condition between two different dielectric media in magnetic field using SCILAB.

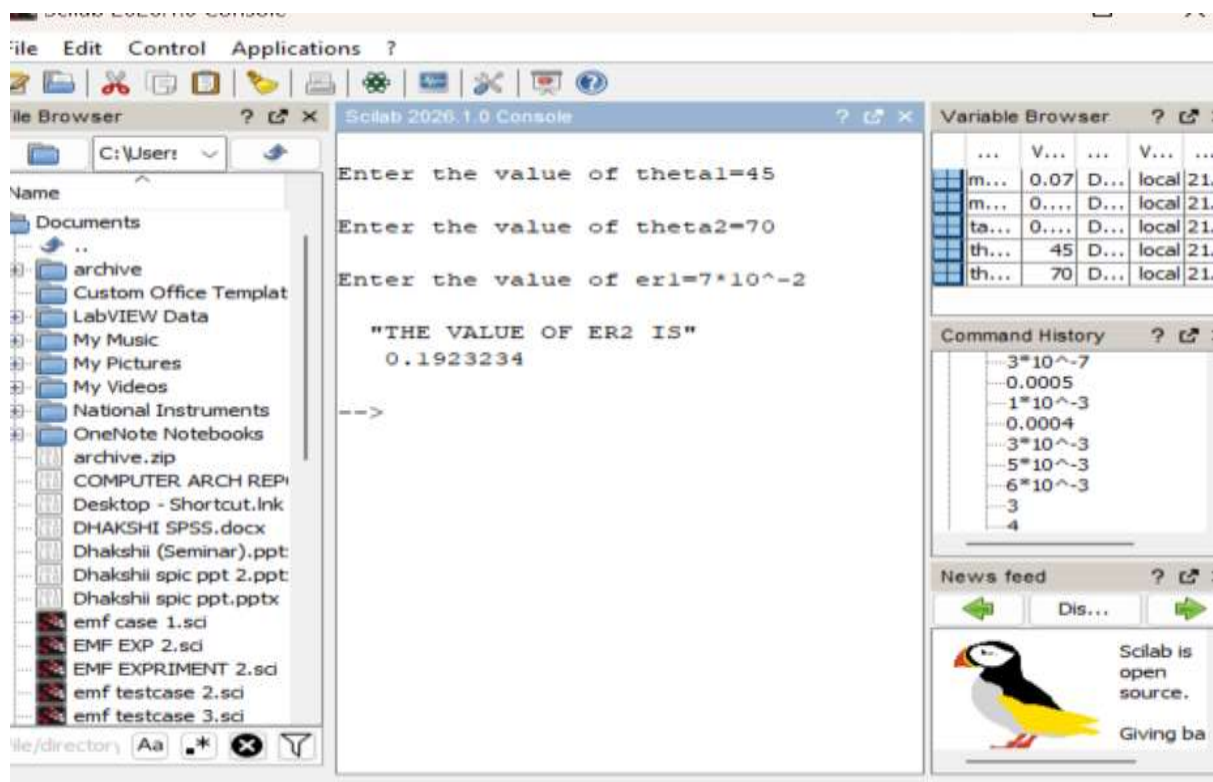
Software required:

- SCILAB version 6.0.1

Partical calucation:



```
1 clc;
2 clear;
3
4 theta_1=input("Enter the value of theta1=");
5 theta_2=input("Enter the value of theta2=");
6 medium_1=input("Enter the value of er1=");
7
8 tan_theta= tand(theta_1)/tand(theta_2);
9 medium_2= medium_1/tan_theta;
10 disp('THE VALUE OF ER2 IS',[medium_2]);
```



Theoretical calucation:

$$\begin{aligned}
 \theta_1 &= 60^\circ = \sqrt{3} = 1.732 \\
 \theta_2 &= 120^\circ = -\tan(180-60) = -\tan 60^\circ = -1.732 \\
 m_1 &= 8 \\
 \tan \theta &= \frac{m_2 - m_1}{\tan \theta_2 - \tan \theta_1} \\
 &= \frac{1.732}{-1.732} = -1 \\
 \frac{m_2 - m_1}{\tan \theta} &= \frac{8}{-1} = -8
 \end{aligned}$$

RESULT:

Thus the effect of two different media on the magnetic field and flux density across the boundary is calculated.